Complete ablation landscape

42 configurations · seeds 120–129 · five-fold predictor per seed

	Pearson r ↑	Kendall τ ↑	MAE10 ↓	RSE ↓
Reference				
TIGER-pair	0.654	0.473	0.381	0.666
Feature modalities				
P + G	0.593	0.426	0.403	0.746
P + S	0.552	0.403	0.408	0.785
S + G	0.532	0.384	0.406	0.769
P only	0.532	0.392	0.405	0.771
G only	0.500	0.360	0.422	0.803
S only	0.056	0.035	0.491	1.106
Graph encoder				
GAT	0.615	0.442	0.389	0.700
GraphSAGE	0.609	0.438	0.383	0.681
GCN	0.567	0.407	0.404	0.741
Node attributes				
+ Energy	0.550	0.393	0.405	0.770
+ Coordinates	0.448	0.316	0.418	0.830
+ Energy + coordinates	0.536	0.384	0.409	0.792
+ Learned embedding	0.600	0.428	0.389	0.709
Zero node features	0.531	0.387	0.407	0.781
Edge geometry				
Sequence-chain edges	0.632	0.451	0.387	0.689
Feature fusion				
Cross-attention: P query	0.630	0.450	0.390	0.699
Cross-attention: G query	0.559	0.400	0.395	0.730
Feature concatenation	0.604	0.426	0.405	0.757
Self-attention: G/S/P	0.595	0.429	0.394	0.717
Self-attention: S/G/P	0.605	0.439	0.391	0.707
Prediction design				
Concatenated pair head	0.601	0.435	0.391	0.707
Single-peptide regression	0.627	0.451	0.402	0.752
Similarity gate				
Training gate off	0.438	0.300	0.425	0.833
Similarity ≥ 0.0	0.451	0.311	0.421	0.823
Similarity ≥ 0.3	0.463	0.329	0.411	0.785
Similarity ≥ 0.5	0.596	0.427	0.393	0.706
Similarity ≥ 0.9	0.619	0.448	0.394	0.711
Signed pair quota				
Signed quota 1000	0.618	0.452	0.399	0.733
Signed quota 2000	0.644	0.467	0.380	0.668
Signed quota 4000	0.641	0.464	0.381	0.668
Signed quota 8000	0.641	0.465	0.384	0.680
Signed quota 10000	0.636	0.463	0.383	0.676
Signed quota 20000	0.634	0.463	0.381	0.671
Unsigned pair quota				
Unsigned quota 1000	0.645	0.465	0.385	0.683
Unsigned quota 2000	0.623	0.451	0.388	0.695
Unsigned quota 4000	0.636	0.460	0.384	0.676
Unsigned quota 8000	0.639	0.458	0.385	0.680
Unsigned quota 10000	0.653	0.474	0.380	0.663
Training / validation pairs				
Negative-pair strategy: top-k	0.619	0.455	0.390	0.701
Strict within-fold validation	0.647	0.467	0.384	0.679
Negative-sample exposure				
P00: neither stage	0.548	0.399	0.409	0.778



Relative performance within each metric

Lower — Higher

Cells: raw means.
Colour: per-metric scale.